

Lab 1

1. Create your tables with their columns in PostgreSQL.

-- Create 'track' table

```
CREATE TABLE track (  
    id SERIAL PRIMARY KEY,  
    track_name VARCHAR(100) NOT NULL  
);
```

-- Create 'subject' table

```
CREATE TABLE subject (  
    id SERIAL PRIMARY KEY,  
    sub_name VARCHAR(100) NOT NULL,  
    max_score INTEGER NOT NULL  
);
```

-- Create 'exam' table

```
CREATE TABLE exam (  
    id SERIAL PRIMARY KEY,  
    date DATE NOT NULL  
);
```

-- Create 'student' table

```
CREATE TABLE student (  
    id SERIAL PRIMARY KEY,  
    e_name VARCHAR(100) NOT NULL,  
    email VARCHAR(150) UNIQUE NOT NULL,  
    address VARCHAR(200),  
    track_id INTEGER REFERENCES track(id) ON DELETE SET NULL,  
    birth_date DATE,  
    gender VARCHAR(6) CHECK (gender IN ('Male', 'Female'))  
);
```

-- Create 'grades' table

```
CREATE TABLE grades (  
    stu_id INTEGER,  
    sub_id INTEGER,  
    exam_id INTEGER,  
    grade NUMERIC(5,2),  
    PRIMARY KEY (stu_id, sub_id, exam_id),  
    FOREIGN KEY (stu_id) REFERENCES student(id) ON DELETE CASCADE,  
    FOREIGN KEY (sub_id) REFERENCES subject(id) ON DELETE CASCADE,  
    FOREIGN KEY (exam_id) REFERENCES exam(id) ON DELETE CASCADE  
);
```

-- Create 'stu_sub' table (student-subject linking table)

```
CREATE TABLE stu_sub (  
    stu_id INTEGER,
```

```
sub_id INTEGER,  
PRIMARY KEY (stu_id, sub_id),  
FOREIGN KEY (stu_id) REFERENCES student(id) ON DELETE CASCADE,  
FOREIGN KEY (sub_id) REFERENCES subject(id) ON DELETE CASCADE  
);
```

-- Create 'track_sub' table (track-subject linking table)

```
CREATE TABLE track_sub (  
    track_id INTEGER,  
    sub_id INTEGER,  
    PRIMARY KEY (track_id, sub_id),  
    FOREIGN KEY (track_id) REFERENCES track(id) ON DELETE CASCADE,  
    FOREIGN KEY (sub_id) REFERENCES subject(id) ON DELETE CASCADE  
);
```

```
alaa@lol:~/Desktop$ sudo -i -u postgres
[sudo] password for alaa:
postgres@lol:~$ psql
psql (14.17 (Ubuntu 14.17-0ubuntu0.22.04.1))
Type "help" for help.

postgres=# CREATE DATABASE advpostgres;
CREATE DATABASE
postgres=# \c advpostgres
You are now connected to database "advpostgres" as user "postgres".
advpostgres=#
advpostgres=# -- Create 'track' table
CREATE TABLE track (
  id SERIAL PRIMARY KEY,
  track_name VARCHAR(100) NOT NULL
);
-- Create 'subject' table
CREATE TABLE subject (
  id SERIAL PRIMARY KEY,
  sub_name VARCHAR(100) NOT NULL,
  max_score INTEGER NOT NULL
);
-- Create 'exam' table
CREATE TABLE exam (
  id SERIAL PRIMARY KEY,
  date DATE NOT NULL
);
-- Create 'student' table
CREATE TABLE student (
  id SERIAL PRIMARY KEY,
  e_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE NOT NULL,
  address VARCHAR(200),
  track_id INTEGER REFERENCES track(id) ON DELETE SET NULL,
  birth_date DATE,
  gender VARCHAR(6) CHECK (gender IN ('Male', 'Female'))
);
-- Create 'grades' table
CREATE TABLE grades (
  stu_id INTEGER,
  sub_id INTEGER,
```

```

  id SERIAL PRIMARY KEY,
  sub_name VARCHAR(100) NOT NULL,
  max_score INTEGER NOT NULL
);
-- Create 'exam' table
CREATE TABLE exam (
  id SERIAL PRIMARY KEY,
  date DATE NOT NULL
);
-- Create 'student' table
CREATE TABLE student (
  id SERIAL PRIMARY KEY,
  e_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE NOT NULL,
  address VARCHAR(200),
  track_id INTEGER REFERENCES track(id) ON DELETE SET NULL,
  birth_date DATE,
  gender VARCHAR(6) CHECK (gender IN ('Male', 'Female'))
);
-- Create 'grades' table
CREATE TABLE grades (
  stu_id INTEGER,
  sub_id INTEGER,
  exam_id INTEGER,
  grade NUMERIC(5,2),
  PRIMARY KEY (stu_id, sub_id, exam_id),
  FOREIGN KEY (stu_id) REFERENCES student(id) ON DELETE CASCADE,
  FOREIGN KEY (sub_id) REFERENCES subject(id) ON DELETE CASCADE,
  FOREIGN KEY (exam_id) REFERENCES exam(id) ON DELETE CASCADE
);
-- Create 'stu_sub' table (student-subject linking table)
CREATE TABLE stu_sub (
  stu_id INTEGER,
  sub_id INTEGER,
  FOREIGN KEY (sub_id) REFERENCES subject(id) ON DELETE CASCADE,
  CREATE TABLE
  CREATE TABLE
  CREATE TABLE
  CREATE TABLE
  CREATE TABLE
  CREATE TABLE
```

2. Insert at minimum 3 Rows at each table.

-- Insert into 'track'

INSERT INTO track (track_name) VALUES
('Web Development'),
('Data Science'),
('Cybersecurity'),
('AI and Machine Learning'),
('Operating Systems');

-- Insert into 'subject'

```
INSERT INTO subject (sub_name, max_score) VALUES
('Databases', 100),
('Networks', 100),
('Programming', 100),
('Machine Learning', 100),
('Operating Systems', 100);
```

```
-- Insert into 'exam'
```

```
INSERT INTO exam (date) VALUES
('2024-05-01'),
('2024-05-05'),
('2024-05-10'),
('2024-05-15'),
('2024-05-20');
```

```
-- Insert into 'student'
```

```
INSERT INTO student (e_name, email, address, track_id, birth_date, gender) VALUES
('Ahmed Youssef', 'ahmed.youssef@gmail.com', 'Cairo', 1, '1999-03-15', 'Male'),
('Sara Mahmoud', 'sara.mahmoud@gmail.com', 'Giza', 2, '2000-07-22', 'Female'),
('Mohamed Ali', 'mohamed.ali@gmail.com', 'Alexandria', 3, '1998-11-05', 'Male'),
('Nourhan Adel', 'nourhan.adel@gmail.com', 'Mansoura', 4, '2001-02-17', 'Female'),
('Khaled Saeed', 'khaled.saeed@gmail.com', 'Tanta', 5, '1997-09-30', 'Male');
```

```
-- Insert into 'grades'
```

```
INSERT INTO grades (stu_id, sub_id, exam_id, grade) VALUES
(1, 1, 1, 95.5),
(2, 2, 2, 88.0),
(3, 3, 3, 76.5),
(4, 4, 4, 91.0),
(5, 5, 5, 82.0);
```

```
-- Insert into 'stu_sub'
```

```
INSERT INTO stu_sub (stu_id, sub_id) VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);
```

```
-- Insert into 'track_sub'
```

```
INSERT INTO track_sub (track_id, sub_id) VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);
```

```
Activities Terminal 29 Apr 3:01 PM
alaa@lol: ~/Desktop

-- Insert into 'subject'
INSERT INTO subject (sub_name, max_score) VALUES
('Databases', 100),
('Networks', 100),
('Programming', 100),
('Machine Learning', 100),
('Operating Systems', 100);

-- Insert into 'exam'
INSERT INTO exam (date) VALUES
('2024-05-01'),
('2024-05-05'),
('2024-05-10'),
('2024-05-15'),
('2024-05-20');

-- Insert into 'student'
INSERT INTO student (e_name, email, address, track_id, birth_date, gender) VALUES
('Ahmed Youssef', 'ahmed.youssef@gmail.com', 'Cairo', 1, '1999-03-15', 'Male'),
('Sara Mahmoud', 'sara.mahmoud@gmail.com', 'Giza', 2, '2000-07-22', 'Female'),
('Mohamed Ali', 'mohamed.ali@gmail.com', 'Alexandria', 3, '1998-11-05', 'Male'),
('Nourhan Adel', 'nourhan.adel@gmail.com', 'Mansoura', 4, '2001-02-17', 'Female'),
('Khaled Saeed', 'khaled.saeed@gmail.com', 'Tanta', 5, '1997-09-30', 'Male');

-- Insert into 'grades'
INSERT INTO grades (stu_id, sub_id, exam_id, grade) VALUES
(1, 1, 1, 95.5),
(2, 2, 2, 88.0),
(3, 3, 3, 76.5),
(4, 4, 4, 91.0),
(5, 5, 5, 82.0);

-- Insert into 'stu_sub'
INSERT INTO stu_sub (stu_id, sub_id) VALUES
(1, 1),
(2, 2),
(3, 3),
(5, 5);

INSERT INTO track_sub (track_id, sub_id) VALUES
INSERT 0 5
INSERT 0 5
INSERT 0 5
INSERT 0 5
INSERT 0 5
INSERT 0 5
INSERT 0 5
```

```
Activities Terminal 29 Apr 3:01 PM
alaa@lol: ~/Desktop

advpostgres=# -- Insert into 'track'
INSERT INTO track (track_name) VALUES
('Web Development'),
('Data Science'),
('Cybersecurity'),
('AI and Machine Learning'),
('Operating Systems');

-- Insert into 'subject'
INSERT INTO subject (sub_name, max_score) VALUES
('Databases', 100),
('Networks', 100),
('Programming', 100),
('Machine Learning', 100),
('Operating Systems', 100);

-- Insert into 'exam'
INSERT INTO exam (date) VALUES
('2024-05-01'),
('2024-05-05'),
('2024-05-10'),
('2024-05-15'),
('2024-05-20');

-- Insert into 'student'
INSERT INTO student (e_name, email, address, track_id, birth_date, gender) VALUES
('Ahmed Youssef', 'ahmed.youssef@gmail.com', 'Cairo', 1, '1999-03-15', 'Male'),
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('Mohamed Ali', 'mohamed.ali@gmail.com', 'Alexandria', 3, '1998-11-05', 'Male'),
('Nourhan Adel', 'nourhan.adel@gmail.com', 'Mansoura', 4, '2001-02-17', 'Female'),
('Khaled Saeed', 'khaled.saeed@gmail.com', 'Tanta', 5, '1997-09-30', 'Male');

-- Insert into 'grades'
INSERT INTO grades (stu_id, sub_id, exam_id, grade) VALUES
(1, 1, 1, 95.5),
(2, 2, 2, 88.0),
(3, 3, 3, 76.5),
(4, 4, 4, 91.0),
(5, 5, 5, 82.0);

-- Insert into 'stu_sub'
INSERT INTO stu_sub (stu_id, sub_id) VALUES
(1, 1),
(2, 2),
(3, 3),
```

3. Add birth date column for the student table.

```
CREATE TABLE student (
  id SERIAL PRIMARY KEY,
  e_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE NOT NULL,
  address VARCHAR(200),
  track_id INTEGER REFERENCES track(id) ON DELETE SET NULL,
  birth_date DATE,
```

4. Add gender column which hold only 2 values (Male or Female).

```
CREATE TABLE student (
    id SERIAL PRIMARY KEY,
    e_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) UNIQUE NOT NULL,
    address VARCHAR(200),
    track_id INTEGER REFERENCES track(id) ON DELETE SET NULL,
    birth_date DATE,
    gender VARCHAR(6) CHECK (gender IN ('Male', 'Female'))
);
```

5. Add/Alter foreign key constraints in your tables.

```
CREATE TABLE student (
    id SERIAL PRIMARY KEY,
    e_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) UNIQUE NOT NULL,
    address VARCHAR(200),
    track_id INTEGER REFERENCES track(id) ON DELETE SET NULL,
```

```
);

CREATE TABLE grades (
    stu_id INTEGER,
    sub_id INTEGER,
    exam_id INTEGER,
    grade NUMERIC(5,2),
    PRIMARY KEY (stu_id, sub_id, exam_id),
    FOREIGN KEY (stu_id) REFERENCES student(id) ON DELETE CASCADE,
    FOREIGN KEY (sub_id) REFERENCES subject(id) ON DELETE CASCADE,
    FOREIGN KEY (exam_id) REFERENCES exam(id) ON DELETE CASCADE
);
```

6. Display male students who are born before 1991-10-01.

```
advpostgres=# SELECT e_name, email, birth_date, gender
FROM student
WHERE gender = 'Male'
    AND birth_date < '1991-10-01';
 e_name | email | birth_date | gender 
-----+-----+-----+-----
(0 rows)
```

7. Display students' names that begin with A.

```
advpostgres=# SELECT e_name, email
FROM student
WHERE e_name LIKE 'A%';
 e_name | email 
-----+-----
Ahmed Youssef | ahmed.youssef@gmail.com
(1 row)
```

8. Display subjects and their max score sorted by max score.

```
advpostgres=# SELECT sub_name, max_score
FROM subject
ORDER BY max_score DESC;
   sub_name   | max_score
-----+-----
 Databases    |         100
 Networks     |         100
 Programming   |         100
 Machine Learning |         100
 Operating Systems |         100
(5 rows)
```

9. Display the subject with highest max score

```
advpostgres=# SELECT sub_name, max_score
FROM subject
WHERE max_score = (SELECT MAX(max_score) FROM subject);
   sub_name   | max_score
-----+-----
 Databases    |         100
 Networks     |         100
 Programming   |         100
 Machine Learning |         100
 Operating Systems |         100
(5 rows)
```